Tema Esame 10/01/25

In 1877 the Italian astronomer Giovanni Schiaparelli, director of the Brera's observatory in Milano, discovered the Mars' channels using the Merz-Repsold telescope, one of the best instruments of the time. The restored telescope is now visible in the Science and Technology Museum in Milano.

Let's suppose that, using a scientific grade CCD camera, we want to take a picture of the central region of the *full moon* (luna piena in IT) using this old instrument. A few technical data of the telescope and the CCD camera are listed in the table.

Assume that:

i) the distance d of the Sun from the Moon is approximately $d \approx 150 \times 10^6$ km and its radius R is $R \approx 700.000$ km.

ii) the solar irradiance SI on the Moon is equal to that on the Earth, i.e. $SI = 1360$ W/m^2, **40% of which falls within the visible band (assume monochromatic light with $\lambda = 550$ nm);**

iii) the surface of the Moon is approximately lambertian, with an average diffuse reflectance (albedo) $A = 10\%$.

Telescope Parameter	**Value**
Lens diameter	$D = 49$ cm
Focal length	$L = 7$ m
CCD Parameter	**Value**
ϕ_s at inversion thresh.	$\phi_s = 0.7$ V
Oxide capacitance	$Ci = 70$ nF/cm^2
Parameter V_0	$V_0 = 0.4$ V
High Gate Voltage	$V_{GH} = 10$ V
Quantum efficiency	$\eta = 80\%$
Pixel Area	$A_{Pixel} = 25$ µm^2
Fill factor	$Ff = 50\%$

Calculate:

a) the radiance of the Sun;

b) the radiance of the Moon;

c) the irradiance on the CCD sensor;

d) the full well capacity Q_{Max} of CCD's pixels using the fundamental equation of the CCD:
$$V_{GH} = \sqrt{2V_0\phi_s} + \phi_s - Q_M/C_i;$$

e) the exposure time required to fill the CCD's potential well with a fill factor $F_f = 50\%$ of the charge Q_{Max} calculated in point d).

Solution

Question a)

The radiance of the Sun can be calculated using two equivalent methods.

i) Isotropy of the emission and conservation of the energy

The radiant flux emitted by the sun Φ_{Sun} can be calculated by integrating the Solar Constant, i.e. the irradiance at distance d, over a spherical surface of radius d. Then, the exitance of the Sun M_{Sun} can be obtained by dividing Φ_{Sun} by its surface. Finally, the radiance L_{Sun} is given by M_{Sun}/π.

$$\Phi_{Sun} := 4\, SI\, \pi\, d^2 \tag{1}$$

$$M_{Sun} := \frac{\Phi_{Sun}}{4\,\pi\, R^2} \tag{2}$$

$$L_{Sun} := \frac{M_{Sun}}{\pi} \tag{3}$$

$$L_{Sun} = \frac{SI\, d^2}{R^2\, \pi} \tag{3}$$

ii) Irradiance produced by a disk at a large distance along its axis. Using the inverse square law approximation, this method give the very same result of method i).

$$SI = \frac{L_{Sun}\, \pi\, R^2}{d^2} \tag{4}$$

Solving Eq. (4) for L_{Sun} we get the required value.

$$L_{Sun} := \frac{SI\, d^2}{R^2\, \pi} \tag{5}$$

Question b)

a) Radiance of the Moon

The radiance of the Moon is simply obtained by transforming its irradiance (SI) into the exitance through the albedo A and dividing by π.

$$L_{Moon} := \frac{SI\,A}{\pi} \tag{6}$$

The visible radiance is then

$$L_{Moon}^{VIS} := \frac{SI\,A\,VIS}{\pi} \tag{7}$$

b) Irradiance on the CCD sensor

To obtain this quantity we can rely on the quasi-lambertian properties of the Moon surface and on the focal ratio (or F number) of the telescope.

i) F number of the telescope

$$f\# := \frac{L}{D} \tag{8}$$

ii) Irradiance on the CCD in the visible spectral range

Since the maglification ratio is very small, the simple equation $E = \frac{M}{4 f\#^2}$ can be used to calculate the irradiance.

$$E_{CCD} := \frac{\pi\,L_{Moon}\,VIS}{4\,f\#^2} \tag{9}$$

$$E_{CCD} = \frac{SI\,A\,VIS\,D^2}{4\,L^2} \tag{9}$$

c) Full well capacity of the CCD

i) Equation of the CCD

$$V_{GH} = \sqrt{2}\,\sqrt{\phi_s\,V_0} + \phi_s - \frac{Q}{C_i} \tag{10}$$

ii) Maximum charge density set only by vertical confinement

$$Q_{Max} := \left(\sqrt{2}\,\sqrt{\phi_s\,V_0} - V_{GH} + \phi_s\right) C_i \tag{11}$$

iii) Full well capacity of CCD Pixels

$$Q_{Pixel} = \left(\sqrt{2}\,\sqrt{\phi_s\,V_0} - V_{GH} + \phi_s\right) C_i\,A_{Pixel} \tag{12}$$

d) Exposure time

i) Maximum number of electrons per pixel considering the fill factor

$$e_{Max} := -\frac{Q_{Pixel}\,F_f}{e} \tag{13}$$

ii) Responsivity in terms of electrons per watt @ λ

$$R_\lambda := \frac{\eta\,\lambda}{h\,c} \tag{14}$$

iii) Rate of average electron accumulation in each pixel (e/s)

$$e_r := E_{CCD}\,A_{Pixel}\,R_\lambda \tag{15}$$

$$e_r = \frac{SI\,A\,VIS\,D^2\,A_{Pixel}\,\eta\,\lambda}{4\,L^2\,h\,c} \tag{16}$$

iv) Exposure time

$$T := \frac{e_{Max}}{e_r}$$

$$T = -\frac{4\left(\sqrt{2}\,\sqrt{\phi_s}\,\sqrt{V_0} - V_{GH} + \phi_s\right) C_i\,F_f\,L^2\,h\,c}{e\,SI\,A\,VIS\,D^2\,\eta\,\lambda} \tag{17}$$

Numerical Results

a) Radiance of the Sun

$$L_{Sun} = 1.99 \times 10^{7} \frac{\mathrm{W}}{\mathrm{m}^2\,\mathrm{sr}} \tag{18}$$

b) Radiance of the Moon

$$L_{Moon} = 4.33 \times 10^{1} \frac{\mathrm{W}}{\mathrm{m}^2\,\mathrm{sr}} \tag{19}$$

c) Irradiance on the CCD

$$E_{CCD} = 6.66 \times 10^{-2} \frac{\mathrm{W}}{\mathrm{m}^2} \tag{20}$$

d) Full well capacity

i) Maximum charge density

$$Q_{Max} = -5.99 \times 10^{-3} \frac{\mathrm{C}}{\mathrm{m}^2} \tag{21}$$

ii) Maximum number of electrons per pixel

$$e_{Max} = 4.67 \times 10^{5} \tag{22}$$

e) Exposure time

$$T = 127.\ \mathrm{ms} \tag{23}$$